

Xiangyu Ma

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Wuhan, Hubei, China

RESEARCH INTERESTS

Quantum information; Bell nonlocality and multipartite entanglement; Quantum algorithms; Quantum many-body physics.

EDUCATION

• Jilin University

Sept. 2021 – Jun. 2026

B.Eng. in Traffic and Transportation

Changchun, China

◦ GPA: 3.16/4.0; Overall average: 84.39/100

◦ **Physics and Mathematics:** Quantum Mechanics (95.1), Thermodynamics and Statistical Physics (80.2), Calculus (93), Mathematical Modeling (95), University Physics (85.6), Electrodynamics (82.6), Lie Groups and Lie Algebras (95), and Methods of Mathematical Physics II (86.2).

• Jilin University

Sept. 2025 – Jun. 2026

Minor in Quantum Information Science

Changchun, China

◦ **Selected Coursework:** Introduction to Quantum Information, Quantum Optics, Quantum Computing, and Quantum Precision Measurement.

MANUSCRIPT

Xiangyu Ma and Jaeha Lee, "Fixed-Measurement Quantum Realizability in the Tripartite Bell Scenario," *manuscript in preparation for submission* (2026).

RESEARCH EXPERIENCE

• Fixed-Measurement Quantum Correlations and Tripartite Entanglement

July 2025 – Present

University of Tokyo

Supervisor: Prof. Naomichi Hatano; Research Mentor: Asst. Prof. Jaeha Lee

◦ **Exact quantum realizability.** Derived a necessary-and-sufficient criterion for the quantum realizability of all eight three-body full correlators in the tripartite (3, 2, 2) scenario under fixed nondegenerate local projective measurement pairs, and constructed explicit realizing quantum states.

◦ **Entanglement certification.** Developed a symmetry reduction based on measurement-adapted Pauli coordinates and a data-preserving Klein four-group twirl to characterize the exact biseparable full-correlation body, yielding a fixed-measurement criterion for genuine multipartite-entanglement certification.

◦ **Tripartite entanglement structure.** Investigated the relation between the fully-separable–biseparable–GME hierarchy and the complementary GHZ/W SLOCC classification using canonical-state representations, polynomial invariants, and the block and singular-value structure of tripartite correlation tensors.

◦ **Mixed-state boundary analysis.** Analyzed Mermin- and Svetlichny-violation regions for generalized GHZ, W, and GHZ–W–white-noise mixtures through analytical optimization and high-precision numerical scans, identifying measurement-sensitive boundary behavior and branch switching.

• Nuclear Matter Phase Structure Study

June 2023 – Aug. 2023

Institute of Modern Physics, Chinese Academy of Sciences

Supervisor: Hao Qiu

◦ Investigated nuclear-matter phase structure, including quark confinement, quark–gluon plasma, and critical behavior; implemented ROOT-based data-analysis workflows for exploratory hypernuclear reconstruction.

SKILLS

• **Programming and Scientific Computing:** Python (NumPy, SciPy, CVXPY, Matplotlib), C/C++, MATLAB, CERN ROOT

• **Numerical Methods:** Monte Carlo sampling, analytical and numerical parameter optimization, semidefinite programming, and the NPA hierarchy

• **Machine Learning and Data Analysis:** Feedforward neural networks, GRU and LSTM models, clustering, time-series forecasting, and model evaluation

• **Research Tools:** $\text{\LaTeX}$, Git, and COMSOL

HONORS AND AWARDS

• Second-Class Scholarship (Rank 7/74)

Apr. 2026

Jilin University

• College Excellent Student (Rank 7/74)

Apr. 2026

Jilin University

• Provincial Second Prize, National College Students Mathematical Contest in Modeling

Dec. 2023

Chinese Society of Industrial and Applied Mathematics

• Second Prize, China Undergraduate Physics Tournament (CUPT)

Dec. 2022

College of Physics, Jilin University

SELECTED ACADEMIC PROGRAMS

- **Physics of Open Systems: Resonance, Symmetry and Topology** Aug. 2025
University of Tokyo 
 - Topics: Open quantum systems, non-Hermitian physics, and Floquet topology.
- **International Summer School for Graduate Students on Optics** Aug. 2024
Fudan University
 - Topics: Quantum optics, nanophotonics, and nonlinear and ultrafast optics.